

HRISHIKESH BELAGALI

Undergraduate Student

📍 East Lansing

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SUMMARY

Undergraduate computer science major with research interests in quantum algorithms, molecular dynamics, reservoir computing, quantum dynamics, and high performance computing.

SKILLS

Languages: Python, C, C++, $\text{\LaTeX}$

Packages: NumPy, PyTorch, scipy, Qiskit, Cirq, PyEMMA, hyperopt, VMD-Python

Softwares: LAMMPS, OVITO, NAMD, VMD

EDUCATION

Computer Science (B.S)

Expected Graduation: 2027

Current GPA: 4.0

Michigan State University Honors College

EXPERIENCE

2025–

Honors Research Scholar

MSU Honors College

- Conducted research on generalized quantum simulators under the guidance of Professor Ryan LaRose.

2024–

Undergraduate Research Assistant

Vermaas Lab, Plant Research Laboratory

- Investigated the interaction of the chloroplast targeting protein (At4g16060) with thylakoid membranes using molecular dynamics.
- Analyzed the effect of PDLIP-5 (Plasmodesmata Located Protein) on water flow rates through PIP1;4 aquaporin tetramer using grid-steered molecular dynamics.
- Simulated the effects of small molecules like butanol on *Clostridium acetobutylicum* membrane dynamics.

CONFERENCES

2025

PRL, MSU

Aquaporin Water Transit Modulation via Protein Capping

acs.digitellinc.com

Abstract presented at ACS Fall 2025 Conference

2025

PRL, MSU

Evaluating the Interaction of Chloroplast-targeting Protein (At4g16060) with Thylakoid Membrane

Abstract presented at First Great Lakes Plant Science Conference

PROJECTS

Python

Qiskit

Discretized Quantum Adiabatic Evolution

github.com

Implemented a quantum adiabatic algorithm to solve QUBO type problems using the Trotter-Suzuki approximation and quantum Ising model formulation techniques

Python

Qiskit

Quantum Arithmetic Circuits

github.com

Built quantum circuits to perform addition, subtraction and multiplication using the Quantum Fourier Transform.

Python

NumPy

Monte Carlo Integration

github.com

Monte Carlo integration with importance sampling to numerically evaluate complex integrals.

Python

NumPy

Langevin Monte Carlo

github.com

Metropolis-adjusted Langevin algorithm for sampling from intractable probability distributions.

Python

OVITO

pyMOL

Genetic Annealing to Determine Protein Structures

github.com

Monte Carlo Genetic Algorithm to determine protein structures through the optimization of Irback's off-lattice model energy equation ($\text{RMSD} < 3.0$).

Python

Quadratic Unconstrained Binary Optimization

github.com

Stochastic tunneling-enhanced simulated annealing for solving QUBO-type problems.

Python

NumPy

aqua-blue

github.com/pypi.org

Secondary maintainer of aqua-blue, a package building echo state networks.

- Implemented reservoir state time-lag, sparsity and spectral radius to improve prediction accuracy.
- Introduced timezone-aware datetime support for seamless interfacing with data pipelines.
- Demonstrated prediction capability by predicting quantum time evolution of Gaussian wave packets.

Python NumPy hyperopt	aqua-blue-hyperopt Primary maintainer of aqua-blue-hyperopt, a package implementing hyperparameter optimization functionality for aqua-blue models. • Created modular hyperparameter optimization functionality for aqua-blue reservoir computers. • Implemented custom loss function capabilities for flexible model evaluation.	github.com/pypi.org
Python PyTorch pytest	neurop Primary maintainer of neurop, a package implementing neural operators. • Developed Fourier Neural Operator (FNO) architecture for solving PDEs. • Implemented the Fractional Fourier Transform for use in Complex Neural Operators (CoNO). • Built CI/CD pipeline with GitHub Actions for automated testing and deployment.	github.com/pypi.org

OPEN SOURCE CONTRIBUTIONS

QuTiP Python	Quantum Toolbox in Python (QuTiP) Contributed to QuTiP, an open-source software for simulating quantum systems. • Implemented functionality to return eigenkets as an operator in the Qobj class.	github.com
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HONORS AND AWARDS

- Wielenga Research Scholarship, Honors College (2025-2026)
- Dean's List (2024, 2025)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024)
- MSU President's Scholarship (2024)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024)